



MINING DEALER BEST PRACTICE

Off-Highway Trucks - Quick Diagnostic Tool for A/C Failure

Maintenance and
Repair

Application

Component Life
Management

Component
Renewal (CRC)

MARC
Management

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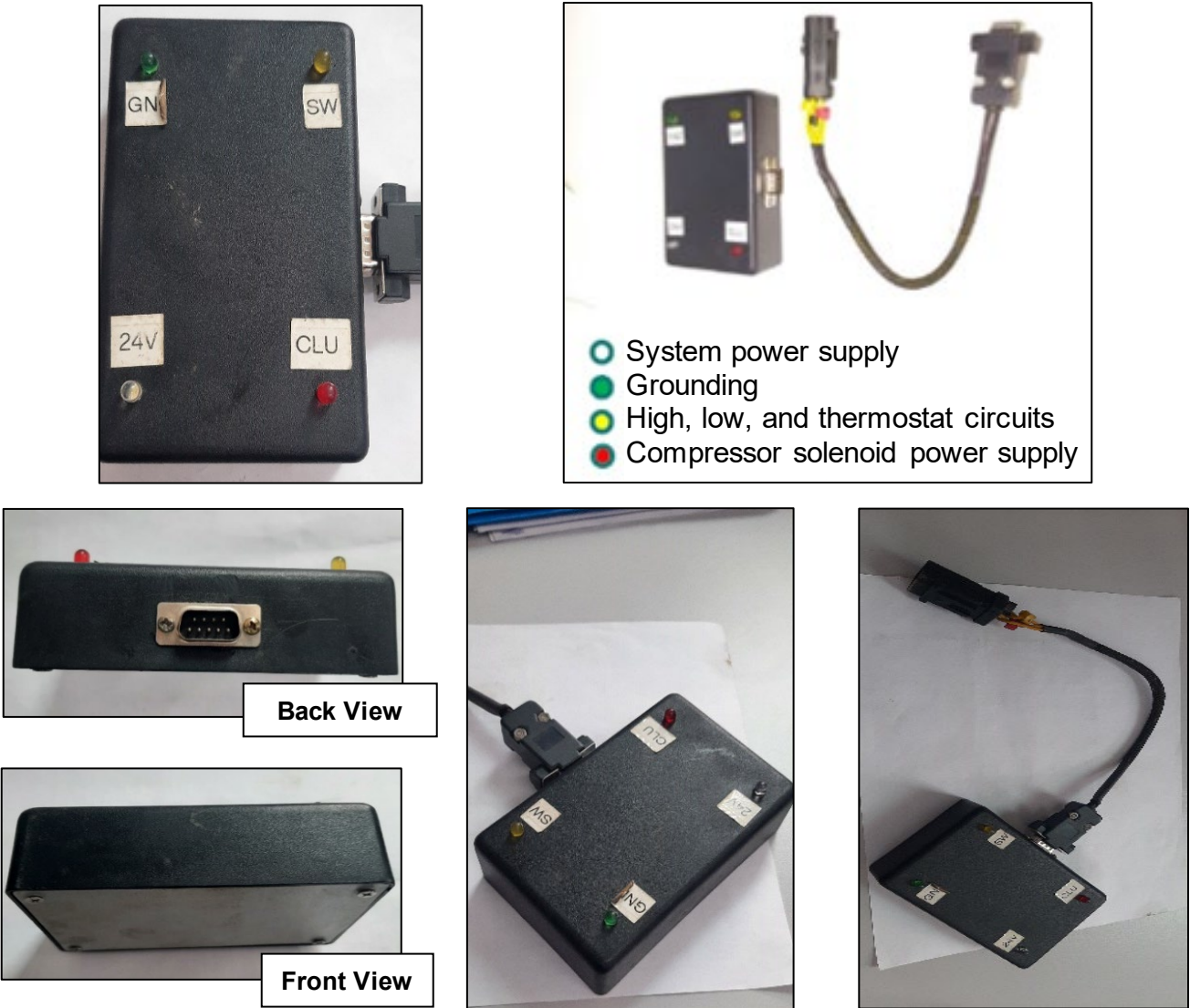
July 2023
1.04-230724-1

1.0 Introduction

The optimization of maintenance activities is often enhanced due to their importance and impact on asset performance. Therefore, the search for special tools or aids to increase equipment physical availability, accuracy, and speed in fault diagnosis is necessary to meet customer expectations. In this scenario, this Best Practice will present a tool option that provides easy and quick diagnostics (approximately within 5 minutes).

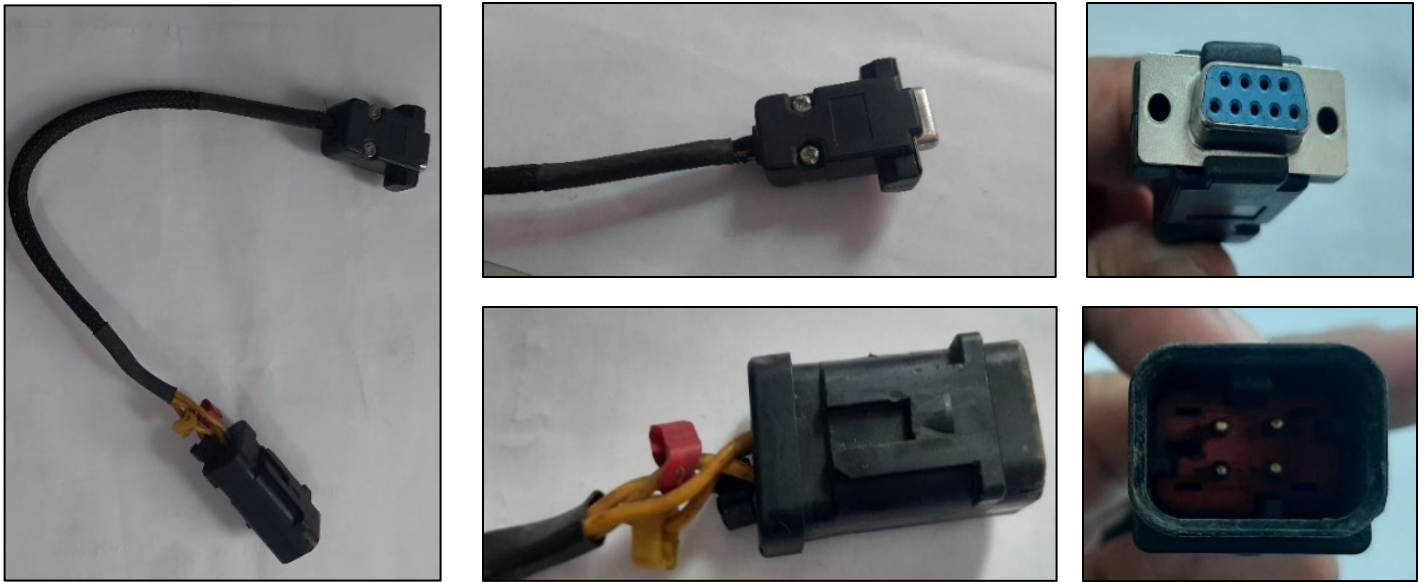
This tool improves fault diagnosis in the air conditioning system, guiding the technician to the damaged circuit:

- Electrical Supply (White **LED 24V** Description).
- Grounding (Green **LED** with **GN** Description).
- Thermostat circuit - high and low circuit (Yellow **LED** with **SW** Description).
- Compressor Clutch Energy Supply (Red **LED** with **CLU** Description).



Picture 1 - device and connection cable.

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Picture 2 - cables and connections present in the device.

2.0 Best Practice Description

Significant delays occur daily in diagnosing air conditioning and electrical systems, due to the numerous checkpoints required to identify the fault. Lack of accuracy and delays in the diagnostic process are recurring issues. The time can range from 2 to 3 hours depending on who is responsible for operating it or the available resources.

The Main Problems Are:

- Insufficient technical resources and equipment to carry out the activities.
- Limited technical knowledge of the teams involved.
- Numerous checkpoints in the air conditioning system.

The charts demonstrate the number of breakdown stops (Chart 1) and the duration of the downtime in hours (Chart 2) for corrective maintenance, highlighting the impact on asset availability and MTBF throughout 2021 related to the air conditioning system:

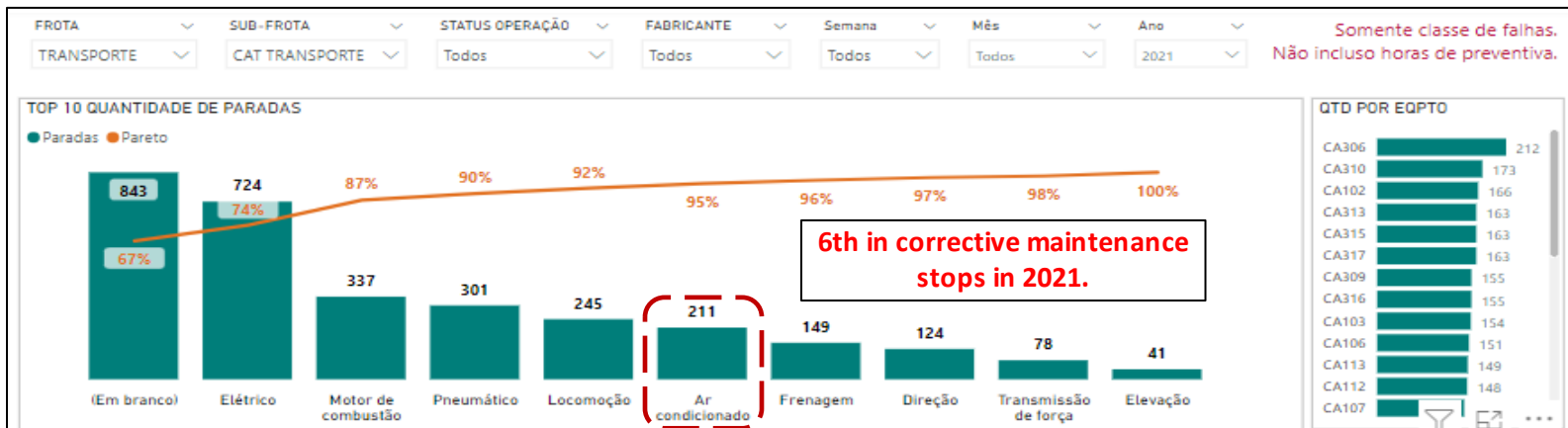


Chart 1 - Number of breakdowns due to corrective maintenance in 2021.

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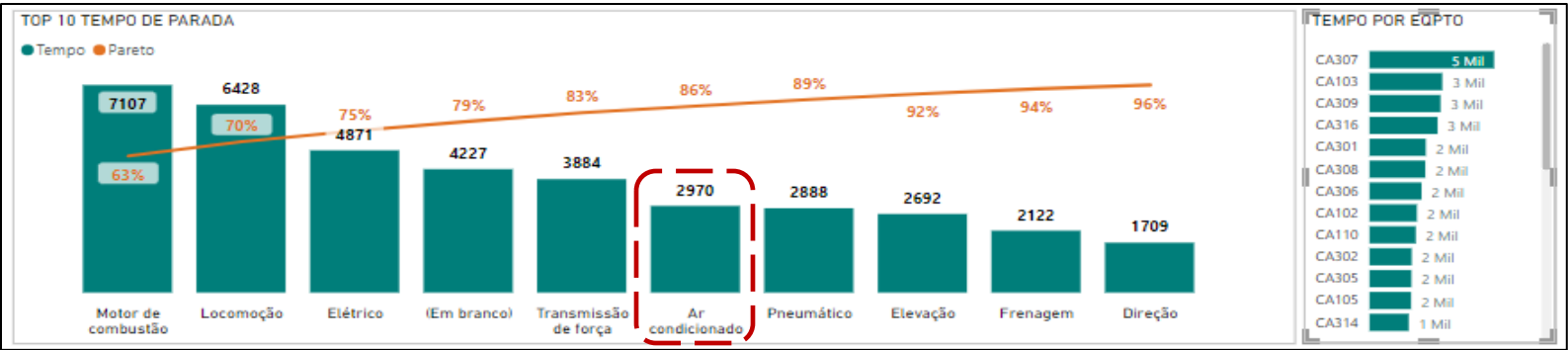
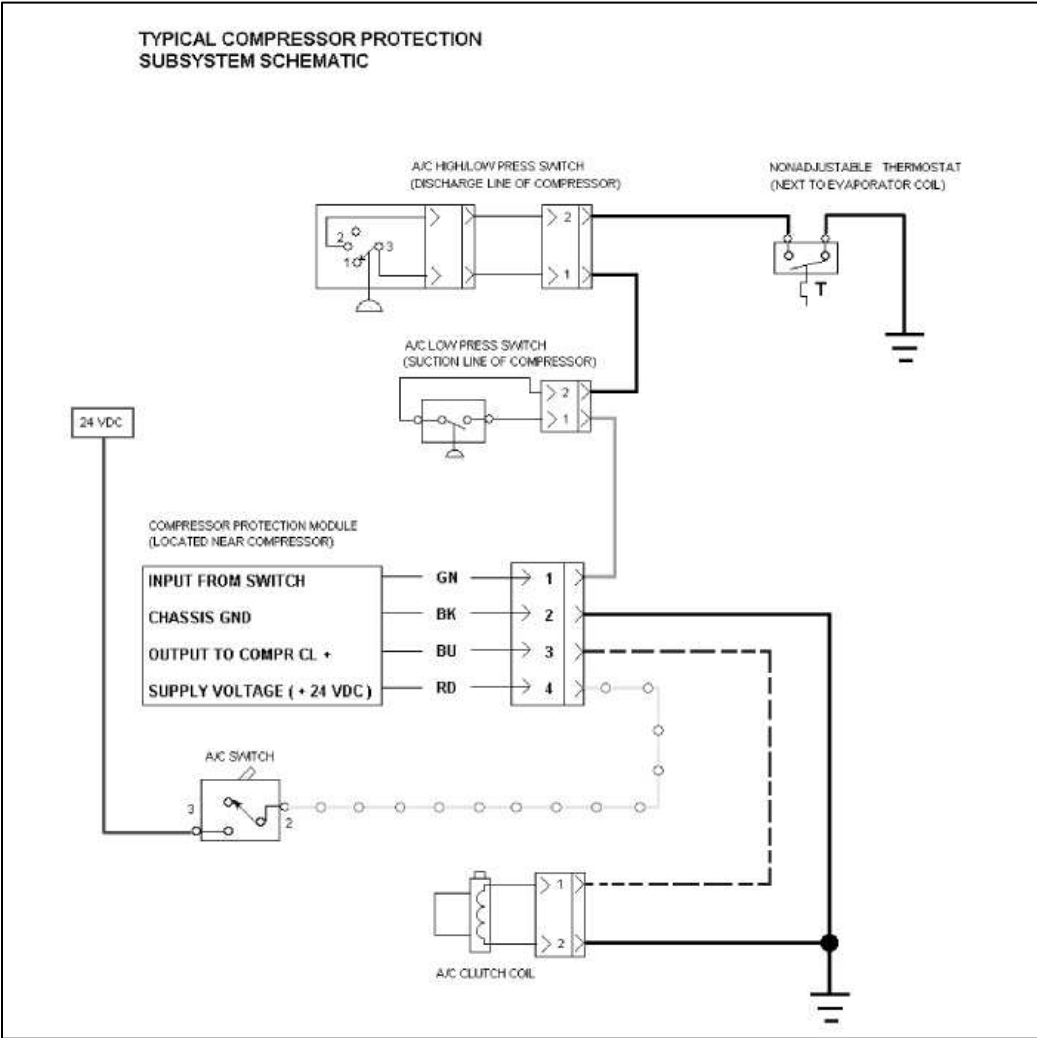


Chart 2 - Total duration of corrective maintenance in 2021.

3.0 Implementation Steps

Next, presentation of the typical compressor protection subsystem schematic:



Picture 3 - Refrigerant Compressor Protection System (SEN5664-35) - i01439520

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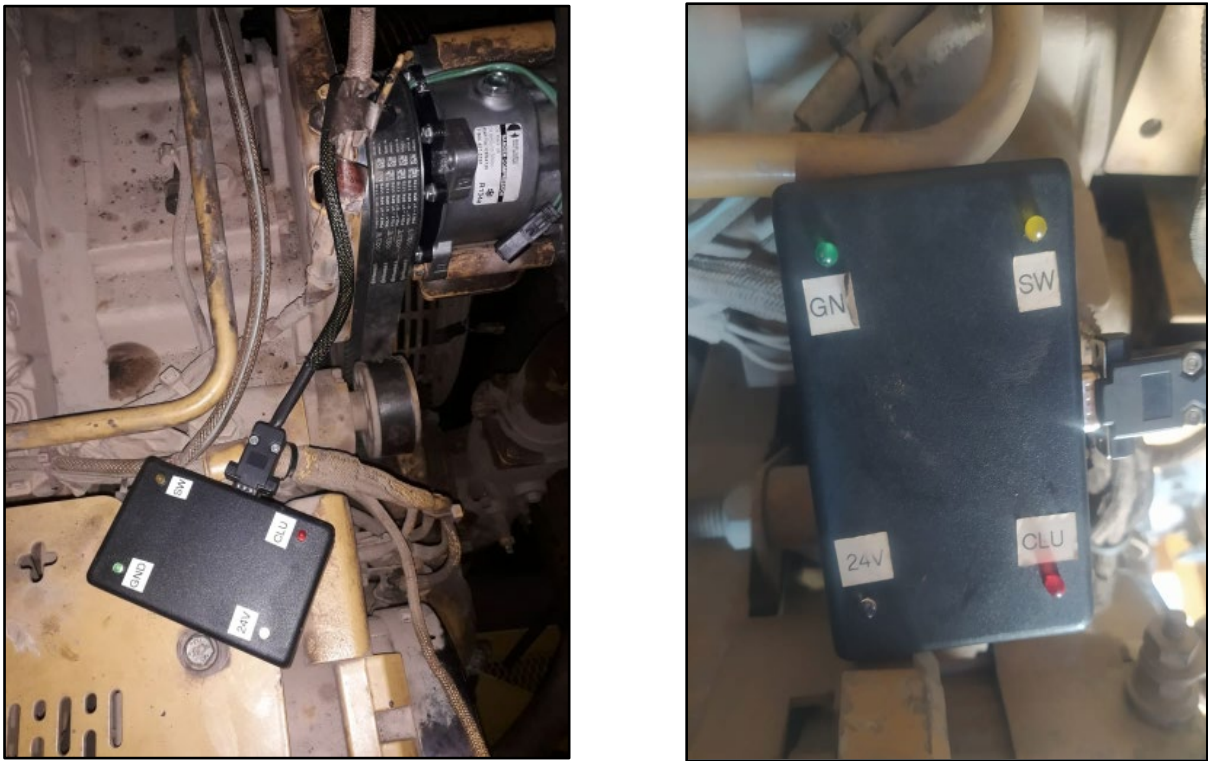
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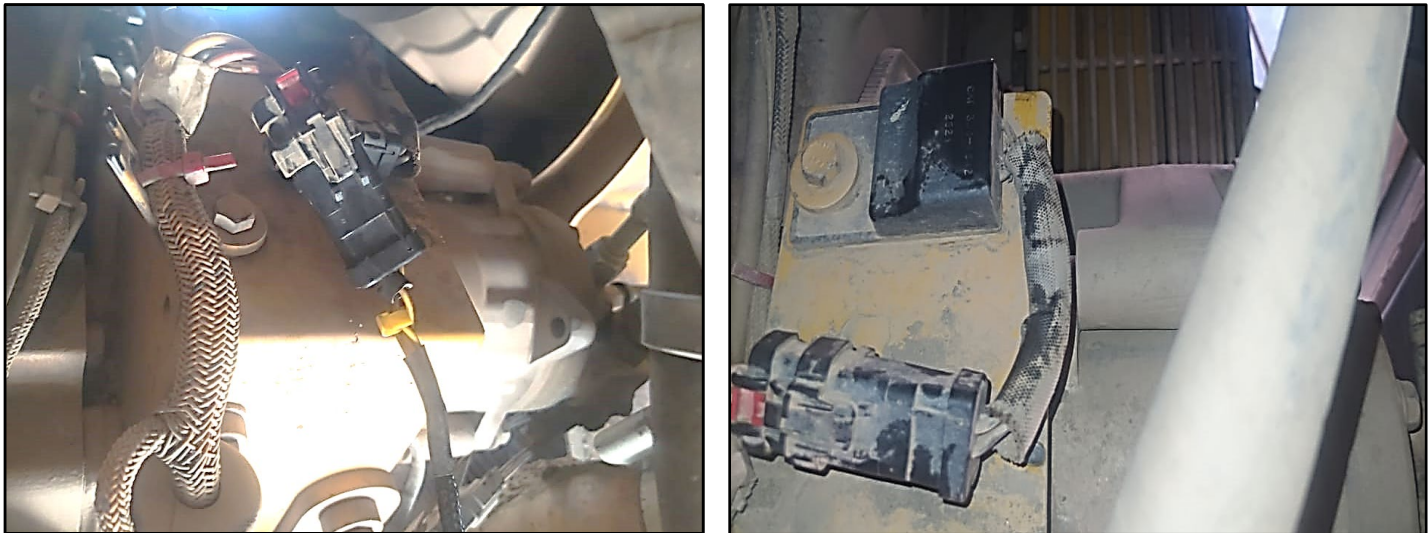
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3.1 Position the device in a place where the LED lights are visible:



Picture 4 - Connection position.

3.2 Connect the Simulator to the **Air-Conditioning** compressor through the connectors as below:



Picture 5 - Compressor connection with cable.

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3.3 Turn on the first stage by the Ignition Key. After power on, the LED lights will turn on (the red LED will turn on after approximately 18 seconds) as below:



Picture 6 - Device in operation.

3.4 If the lights do not turn on, the technician has an indication that there is a problem with the compressor or high/low switch or activation switch, or harness. The picture 7 shows a failure simulation where the high switch circuit was opened (yellow LED with the description SW) indicating a possible failure of the A/C compressor.



Picture 7 - Yellow LED (SW) off.

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4.0 Benefits

- Reduction in the time required to perform air conditioning system fault diagnosis.
- 35% reduction in total corrective maintenance time.
- Easy and cost-effective construction tool.
- The simplicity of use allows technicians with any level of experience or knowledge to use the device.

5.0 Resources Required

QTY	Unity.	Description
01	Unity.	Assembly Case/Box - 40 x 81 x 121 mm.
01	Unity	Nautical Mesh 8mm x 1 (Isolation Cables)
01	Unity	DB9 Connector (Female)
01	Unity	DB9 Connector (Male)
04	Unity	6mm LEDs (Different Colors)
04	Unity	2K2 Resistor
04	Unity	1N4007 diode
02	Unity	LM2904N Integrated Circuit
01	L	Resin for encapsulation
01	Unity.	DNI 0413 Timer Relay
06	Unity.	3/5 x 1" Bolts
01	m	3,0 mm Wire

6.0 Supporting Attachments / References

SIS 2.0 - Refrigerant Compressor Protection System (SENR5664-35) - i01439520.

7.0 Related Best Practices

N/A

8.0 Acknowledgements

Thank you to the mining engineering team at Sotreq S/A.

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